New certified upper bounds for the sign-uncertainty constant and the first autocorrelation constant

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Abstract

We prove two new upper bounds for constants from the AlphaEvolve benchmark collection, both with certificates checked by the kernel of the Lean 4 proof assistant (with Mathlib, no `native_decide`, only the standard axioms). (1) For the one-dimensional sign-uncertainty constant of Bourgain, Clozel and Kahane, $C_{\mathrm{SU}} \le 0.305578$, improving the previous record 0.3089 (and the published bound 0.3216); the witness is $f(x) = P(2\pi x^2)e^{-\pi x^2}$ with P an explicit integer combination of the Laguerre polynomials $L_{2j}^{(-1/2)}$, $j \le 300$. Lean verifies the sign pattern of P exactly; the reduction from the sign pattern to the bound is the classical Hermite eigenfunction argument. (2) For the first autocorrelation (Sidon) constant, $C_1 \le 1.5027434864131$, improving the previous record 1.502743486709583 by $3 \cdot 10^{-10}$; here the continuous statement, including the definition of C_1, is formalised. The kernel checks rely on Kronecker substitution: all coefficients needed by the certificates are read off as base-2^w digits of a few very large integers, which the Lean kernel multiplies with GMP.

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1 Introduction

The 2025 AlphaEvolve reports [13, 9] collected several dozen extremal problems of analysis and combinatorics whose best known bounds are certified by explicit constructions, and since then a number of AI systems and human–AI collaborations have competed on them [1, 14, 5, 2]. In this note we give new upper bounds for two of these constants, both certified in exact arithmetic and by the kernel of the Lean proof assistant.

Sign uncertainty. Bourgain, Clozel and Kahane [4] proved that an even function f with $f(0) \le 0$, $\widehat{f}(0) \le 0$ cannot be eventually nonnegative too early on both sides: $A(f)A(\widehat{f})$ is bounded below by a positive constant. The sharp constant C_{SU} in dimension one is unknown; $0.2025 \le C_{\mathrm{SU}}$ [10], and the upper bound has moved from 0.353 [10] and 0.3284 [8] to 0.3216 (AlphaEvolve), 0.313092 (EinsteinArena) and 0.3089 [5]. We prove $C_{\mathrm{SU}} \le 0.305578$ (Theorem 2.3) with a polynomial of degree 600 in the Gaussian–Laguerre eigenbasis of the Fourier transform. The improvement comes from solving the underlying linear program at a degree well above the double-root families used before, with a numerically robust (log-scaled, cutting-plane) formulation, and from an exact certification method that scales to such degrees.

The first autocorrelation constant. For nonnegative f supported in $[-\frac{1}{4}, \frac{1}{4}]$, $\sup(f * f) \ge C_1(\int f)^2$; $C_1 = 2/\sigma^2$ governs the size of g-Sidon sets [6]. After a long sequence of AI-produced improvements, the record was 1.502743486709583 (EinsteinArena, 13 October 2026). We lower it by $3 \cdot 10^{-10}$ to 1.5027434864131 (Theorem 3.2). The numerical gain is marginal and we do not claim otherwise; the point of including it is that the certificate is, to our knowledge, the first Lean proof of an upper bound for C_1 in which the continuous statement itself (integrals, autoconvolution, the definition of C_1 as an infimum) is formalised, with the 131 071 autoconvolution coefficients of a 65 536-step function checked by a single GMP multiplication in the kernel.

2 The sign uncertainty constant

2.1 The problem

Normalise $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x)e^{-2\pi i x\xi}\,dx$ and, for a real function f, put $A(f) = \inf\{r > 0 : f(x) \ge 0 \text{ for all } |x| \ge r\}$. Let C_{SU} be the largest constant such that $A(f)A(\widehat{f}) \ge C_{\mathrm{SU}}$ for every even, integrable, real f with $\widehat{f}$ integrable and $f(0) < 0$, $\widehat{f}(0) < 0$. This is the one-dimensional "+1" sign uncertainty principle of Bourgain, Clozel and Kahane [4]; Gonçalves, Oliveira e Silva and Steinerberger [10] proved $0.2025 \le C_{\mathrm{SU}} \le 0.353$, Cohn and Gonçalves [8] computed $A_+(1) \le 0.572990$ (so $C_{\mathrm{SU}} \le 0.3284$), and AlphaEvolve [13, 9] obtained 0.3216 (problem B.4 / 6.11, "an uncertainty inequality"). On EinsteinArena [2], whose verifier restricts to at most 25 prescribed double roots, the best value is 0.313092; the multi-agent "Station" of Chung, Du and Wesley [5] reports 0.3089 with a polynomial of degree 226, and shows that families with at most 20 prescribed double roots cannot go below 0.315305.

2.2 Reduction to a polynomial sign condition

Let $L_n^{(\alpha)}$ be the generalised Laguerre polynomials and $\Lambda_n = 2^n n!\, L_n^{(-1/2)} \in \mathbb{Z}[u]$; equivalently $\Lambda_0 = 1$, $\Lambda_1 = 1 - 2u$,

$$\Lambda_{n+2}(u) = (4n + 5 - 2u)\Lambda_{n+1}(u) - 2(n+1)(2n+1)\Lambda_n(u). \tag{2.1}$$

Lemma 2.1 (classical)**.** *For every m, $g_m(x) = \Lambda_m(2\pi x^2)e^{-\pi x^2}$ satisfies $\widehat{g_m} = (-1)^m g_m$.*

Proof. $H_{2m}(y) = (-1)^m 2^{2m} m!\, L_m^{(-1/2)}(y^2)$, and $x \mapsto H_n(\sqrt{2\pi}x)e^{-\pi x^2}$ is an eigenfunction of the Fourier transform with eigenvalue $(-i)^n$ (Hermite functions, rescaled to our normalisation). □

Proposition 2.2. *Let $P = \sum_{j=0}^{D} \gamma_j \Lambda_{2j}$ with real γ_j, and $r > 0$. If $P(0) < 0$ and $P(u) > 0$ for all $u \geq r$, then $C_{\mathrm{SU}} \leq r/(2\pi)$.*

Proof. $f(x) = P(2\pi x^2)e^{-\pi x^2}$ is even, Schwartz, and $\widehat{f} = f$ by Lemma 2.1. Thus $f(0) = \widehat{f}(0) = P(0) < 0$, and $f(x) > 0$ whenever $2\pi x^2 \geq r$, so $A(f)A(\widehat{f}) = A(f)^2 \leq r/(2\pi)$. □

This is exactly the reduction used by GOS, Cohn–Gonçalves, AlphaEvolve and the arena verifier.

2.3 Search

For fixed D and r the existence of P is a linear feasibility problem in (γ_j). We solve it with HiGHS by a cutting-plane method: impose $P(u)e^{-u/2} \geq t\, w(u)$ (with $P(0) = -1$, $t \leq 1$ and a row-normalising weight w) on a finite set of points of $[r, U]$, maximise t, evaluate on a fine grid, add the violated local minima and repeat; the basis functions are evaluated through the stable three-term recurrence for $L_n^{(-1/2)}(u)e^{-u/2}$. Bisection in r gives the optimal radius $r^*(D)$ of the degree-$2D$ family.

2.4 Exact certification of the sign condition

The LP solution is converted to exact binary rationals and then to integers γ_j (common denominator, divided by the gcd). From γ we compute the monomial coefficients of $P = \sum_j \gamma_j \Lambda_{2j}$ by (2.1), choose a rational $r = a/b$ slightly above the optimal radius, and form the integer polynomial

$$R(y) = b^{\deg P}\, P((a+y)/b), \qquad \text{so that} \quad P(u) > 0\ \forall u \geq r \iff R(y) > 0\ \forall y \geq 0.$$

Positivity of R on $[0, \infty)$ is certified by a finite list of *leaves* $(u_0, u_1, v_0, v_1) \in \mathbb{N}^4$, each standing for the interval $\{(u_0 + u_1 z)/(v_0 + v_1 z) : z \geq 0\} = [u_0/v_0, u_1/v_1)$ (or $[u_0/v_0, \infty)$ if $v_1 = 0$). Consecutive leaves share endpoints, the first starts at 0 and the last is unbounded, so the leaves cover $[0, \infty)$. For a leaf put

$$T(z) = \sum_{i=0}^{d} R_i\, (u_0 + u_1 z)^i (v_0 + v_1 z)^{d-i} = (v_0 + v_1 z)^d\, R\Big(\frac{u_0 + u_1 z}{v_0 + v_1 z}\Big).$$

If all coefficients of T are nonnegative and $T(0) > 0$, then $R > 0$ on the leaf (Descartes' rule in its trivial direction). The leaves are produced by the Vincent–Collins–Akritas bisection of $[0, B_0]$ for a dyadic B_0 beyond which $R(B_0 + z)$ has positive coefficients. To check a leaf, the kernel does *not* expand T: it evaluates $N = T(2^w)$ by a homogeneous Horner scheme in big integers (d steps), computes the bound $W = \sum_i |R_i|(u_0 + u_1)^i (v_0 + v_1)^{d-i} \geq \max_j |t_j|$, checks $2W < 2^w$, and reads the coefficients t_j of T off the balanced base-2^w digits of N (Section 4.1). The bound $|t_j| \leq W$ is proved in Lean by a coefficientwise domination argument for the list operations that build T.

2.5 Results

Table 1 lists the optimal radii $r^*(D)$ of the degree-$2D$ family found by the LP (floating point, bisection to 10^{-4}–10^{-6}; for $D \geq 170$ with an interior-point solver); they are *not* certified, only the final instance is. For the certificate we took $D = 300$ and the rational radius $r = 48/25$ slightly

D	$r^*(D)$	$r^*(D)/2\pi$
20	2.0599884	0.3278573
30	1.9989685	0.3181457
40	1.9889709	0.3165546
50	1.9672241	0.3130934
60	1.9670258	0.3130619
70	1.9490326	0.3101982
80	1.9484833	0.3101107
90	1.9484375	0.3101035
100	1.9370929	0.3082979
140	1.9295691	0.3071005
170	1.9255800	0.3064656
200	1.9250488	0.3063810
230	1.9216406	0.3058386
260	1.9206396	0.3056793
300	1.9194482	0.3054897
360	1.9170898	0.3051143

Table 1: Numerical optimal radii for $P = \sum_{j \leq D} \gamma_j \Lambda_{2j}$.

above $r^*(300)$, solved the LP with margin at this r, and rounded.

Theorem 2.3. *Let $P = \sum_{j=0}^{300} \gamma_j \Lambda_{2j}$ with the integers γ_j of `data/su_gamma.txt` (a polynomial of degree 600). Then $P(0) < 0$ and $P(u) > 0$ for all real $u \geq 48/25$. Consequently*

$$C_{\mathrm{SU}} \;\leq\; \frac{48}{2\pi \cdot 25} \;=\; 0.305577490736\ldots < 0.305578.$$

The certificate consists of 41 leaves; the Kronecker base is 2^{16623}; the largest $|\gamma_j|$ has 5342 bits and the largest coefficient of R has 8125 bits. In terms of the constant $A_+(1)$ of [8, 3], $A_+(1) \leq \sqrt{r/2\pi} = 0.55279064$.

3 The first autocorrelation constant

3.1 The constant and the reduction to a finite computation

For $f\colon \mathbb{R} \to \mathbb{R}$ write $(f * f)(t) = \int_{\mathbb{R}} f(x) f(t-x)\, dx$. Let

$$C_1 = \inf\Big\{ \frac{\sup_t (f*f)(t)}{(\int f)^2} \;:\; f \geq 0,\ \operatorname{supp} f \subseteq [-\tfrac{1}{4}, \tfrac{1}{4}],\ \int f > 0 \Big\}.$$

Then $C_1 = 2/\sigma^2$, where σ is the constant governing the maximal size $\sigma\sqrt{gN}(1 + o(1))$ of g-Sidon sets in $\{1, \ldots, N\}$ [6, 7]. The best lower bound is $C_1 \geq 1.2937$ [11] (previously 1.28 [7]). Upper

bounds come from explicit functions: 1.50992 [12], 1.5053 and 1.5032 (AlphaEvolve [13, 9]), 1.50314 [1], 1.50286 [14], and on the EinsteinArena leaderboard [2] a sequence of AI agents pushed it to 1.502743486709583 (agent "Atris", 13 October 2026, a step function with 65536 steps).

Lemma 3.1. *Let $n \geq 1$, $h = 1/(2n)$, $a_0, \dots, a_{n-1} \geq 0$, and $f = \sum_i a_i \mathbf{1}_{[-1/4+ih,\, -1/4+(i+1)h)}$. Put $c_k = \sum_{i+j=k} a_i a_j$. Then $\int f = h \sum_i a_i$ and $(f * f)(t) \leq h \max_k c_k$ for all t. In particular $C_1 \leq 2n \max_k c_k / (\sum_i a_i)^2$.*

Proof. $(f*f)(t) = \sum_{i,j} a_i a_j \, |I_i \cap (t - I_j)|$, and $|I_i \cap (t-I_j)| = h\,\varphi(s-(i+j))$ with $s = (t+\frac{1}{2})/h - 1$ and $\varphi(u) = \max(0, 1-|u|)$. Hence $(f*f)(t) = h \sum_k c_k \varphi(s-k) \leq h \max_k c_k \sum_k \varphi(s-k) \leq h \max_k c_k$, since at most two consecutive integers k have $\varphi(s-k) > 0$ and their weights sum to at most 1. □

(The inequality is in fact an equality at $t = -\frac{1}{2} + (k+1)h$; the arena score is exactly this quantity.)

3.2 Construction

Starting from the public record vector we minimised the log-sum-exp smoothing $\beta^{-1} \log \sum_k e^{\beta c_k / \bar{c}}$ with L-BFGS-B over $a \geq 0$ (fixed $n = 65536$), increasing β up to $2.4 \cdot 10^{10}$ and accepting an iterate only if the *true* objective decreased. The record vector is a highly degenerate minimax point (about 31 000 of the 131 071 values c_k agree with the maximum to relative precision 10^{-9}), and the descent stalls at the limit of double precision. Rounding to integers $0 \leq a_i \leq 2^{40}$ gives the certificate.

Theorem 3.2. $C_1 \leq 15027434864131/10^{13} = 1.5027434864131$. *More precisely, the step function of `data/c1_heights.npy` has $2n \max_k c_k / (\sum a_i)^2 = 1.502743486413064\ldots$*

This improves the previous record by $3.0 \cdot 10^{-10}$: a marginal improvement, which we report because it is certified end to end in Lean (Section 4.2).

4 Formal verification

Both certificates are checked by the kernel of Lean 4 (version 4.35.0) with Mathlib. All computations are performed by `decide +kernel`; no `native_decide`, `sorry` or custom axiom is used, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. The Lean kernel performs arithmetic on natural-number literals with GMP; the certificates are designed so that the expensive part of each check is a small number of operations on very large integers.

4.1 Kronecker substitution

If $a_0, \dots, a_{n-1} \in \mathbb{N}$ and $P = \sum_i a_i 2^{wi}$, then $P^2 = \sum_k c_k 2^{wk}$ with $c_k = \sum_{i+j=k} a_i a_j$; if every $c_k < 2^w$ these are exactly the base-2^w digits of P^2. Signed coefficients are handled by an offset: if $N = \sum_j t_j B^j$ with $|t_j| < B/2$, $B = 2^w$, then the digits of $N + \frac{B}{2} \sum_{j<L} B^j$ are $t_j + \frac{B}{2}$. Lean proves these digit lemmas once (`dig_sum`); all digits of a number with L base-2^w digits are then inspected by a divide-and-conquer Boolean function (`chkMax`, `chkMin`), whose correctness is proved by induction.

4.2 The C_1 certificate (`lean/C1.lean`)

The file contains the general lemmas (Kronecker substitution; the step-function lemma of Section 3 in measure-theoretic form; `cert_implies`, which turns four kernel-checked facts about the packed number P into the continuous statement), the packed data (128 hexadecimal chunks), and the main statements:

```
def AutocorrAdmissible (f : ℝ → ℝ) : Prop :=
  (∀ x, 0 ≤ f x) ∧ (∀ x, x ∉ Set.Icc (-1/4 : ℝ) (1/4) → f x = 0) ∧ Integrable f ∧ 0 < ∫ x, f x

noncomputable def C₁ : ℝ :=
  sInf {c : ℝ | ∃ f : ℝ → ℝ, AutocorrAdmissible f ∧
          ∀ t : ℝ, ∫ x, f x * f (t - x) ≤ c * (∫ x, f x) ^ 2}

theorem C₁_le : C₁ ≤ (15027434864131 : ℝ) / 10000000000000
```

The kernel facts are: $P < (2^{104})^{65536}$; all 65536 digits of P are $\le 2^{40}$; all 131072 digits of P^2 are $\le M$; and digit 65535 of $P \cdot \frac{2^{104\cdot 65536}-1}{2^{104}-1}$ equals $S = \sum_i a_i$, together with $2nMq \le pS^2$. `#print axioms C₁_le` gives `[propext, Classical.choice, Quot.sound]`; the check takes about 100 s and 6 GB.

Audit. The Lean integral of $x \mapsto f(x)f(t-x)$ is the Lebesgue (Bochner) integral, i.e. $(f*f)(t)$; the set in `sInf` is bounded below by 0 (proved), so C_1 is the usual $\inf_f \sup_t (f*f)(t)/(\int f)^2$. The data enter only as a witness.

4.3 The C_{SU} certificate (`lean/su_modules/`)

Because the Lean kernel keeps the intermediate big integers of all leaf checks of a file alive, the certificate is split into modules so that every process stays below 30 GB: `SUCore.lean` (library, data, and the small kernel checks), `SULeaves0`–`SULeaves6.lean` (6 leaf checks each, at most 28 GB and 2 minutes per module on the compute node), and `SUMain.lean`, which imports them and proves the main statement

```
noncomputable def Lam : ℕ → ℝ → ℝ
  | 0, _ => 1
  | 1, u => 1 - 2 * u
  | n + 2, u => (4 * (n : ℝ) + 5 - 2 * u) * Lam (n + 1) u
                - 2 * ((n : ℝ) + 1) * (2 * n + 1) * Lam n u

theorem P_sign :
    (∑ j ∈ Finset.range 301, (γ.getD j 0 : ℝ) * Lam (2 * j) 0) < 0 ∧
    ∀ u : ℝ, (48 : ℝ) / 25 ≤ u →
      0 < ∑ j ∈ Finset.range 301, (γ.getD j 0 : ℝ) * Lam (2 * j) u
```

where γ is the explicit list of integers. The proof uses four kernel checks: `hB` and `hE` (the Kronecker test that the monomial list `P1` is the expansion of $\sum_j \gamma_j \Lambda_{2j}$, Section 2.4), `P_cert` (the chain of leaves and the Kronecker digit test for each leaf, one theorem `hleaf`k per leaf), and `P0_neg` (the constant coefficient of P is negative). `#print axioms SU.P_sign` gives `[propext, Classical.choice, Quot.sound]`; the axiom check follows all imported declarations. A single-file version of the same declarations is provided as well; it needs more than 45 GB of memory and was run only in the modular form.

Audit. `Lam n` is $\Lambda_n = 2^n n!\, L_n^{(-1/2)}$ by the standard Laguerre recurrence $(n+1)L_{n+1}^{(\alpha)} = (2n+1+\alpha-u)L_n^{(\alpha)} - (n+\alpha)L_{n-1}^{(\alpha)}$ with $\alpha = -\frac{1}{2}$; hence the polynomial in `P_sign` is in the span

of $L_{2j}^{(-1/2)}$, which is what Proposition 2.2 needs. The step from `P_sign` to $C_{\mathrm{SU}} \le r/2\pi$ is not formalised.

5 What is and is not verified

- *Lean-checked (kernel, standard axioms only):* for C_1, the full continuous statement (Theorem 3.2, including the definition of C_1 as an infimum and the measure-theoretic computation of the autoconvolution of a step function); for C_{SU}, the sign statement $P(0) < 0$, $P(u) > 0$ for $u \ge r$, for the explicit $P = \sum_j \gamma_j \Lambda_{2j}$ with Λ_n defined by the recurrence (2.1).

- *Paper only:* Lemma 2.1 (Hermite functions are Fourier eigenfunctions, and $\Lambda_n = 2^n n! L_n^{(-1/2)}$) and Proposition 2.2; both are classical and are exactly the reduction used by all previous upper bounds for C_{SU}, including the AlphaEvolve and EinsteinArena verifiers. We also checked the eigenfunction identity numerically for small n.

- *Independent checks:* the C_1 certificate was recomputed by exact `int64` convolutions of 20-bit halves (no Kronecker substitution); the C_{SU} polynomial was checked by a Sturm-sequence real-root count (`sympy`) in addition to the Descartes/VCA certificate.

- *Novelty:* to our knowledge, as of October 14, 2026, the previous best bounds are those listed in the introduction; the field is moving fast (several AI systems post results on EinsteinArena daily), and the C_1 improvement in particular is tiny and may already be superseded.

- *Not addressed:* lower bounds. Our numerical radii $r^*(D)$ decrease slowly with the degree and we make no conjecture about the value of C_{SU}.

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